

GATE

CRASH COURSE

CS/IT

Data Structure & Programming

Lecture 03

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Topics *to be covered*

- 1 Linear Queue, Circular Queue
- 2 Question stack & queue
- 3 Queue using stack
- 4





Queue



Queue is a Linear data structure

Insert
or
delete

1. Queue works based on principle of FIFO
first in first out

2. Queue is 2 ended data structure

front : Deletion

rear : Insertion



Queue



$$f = -1$$
$$r = -1$$

Insert Logic

Increment the rear

Insert in new Index

Empty condition

$$f = -1$$



Queue



$$f = \underline{-1}$$

$r = 0$
first Insertion
 $f = 0$



Queue



$f=0$

$r=1$



Queue



$f=0$



$r=2$





Queue



$f=0$

$r=3$

$rear = rear + 1$



Queue

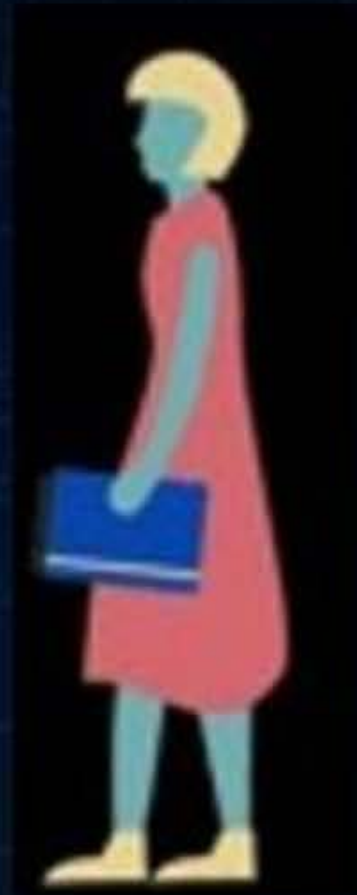


$f=0$

$r=4$



Queue



$f=0$

Deletion
delete element from front
Index $r=5$



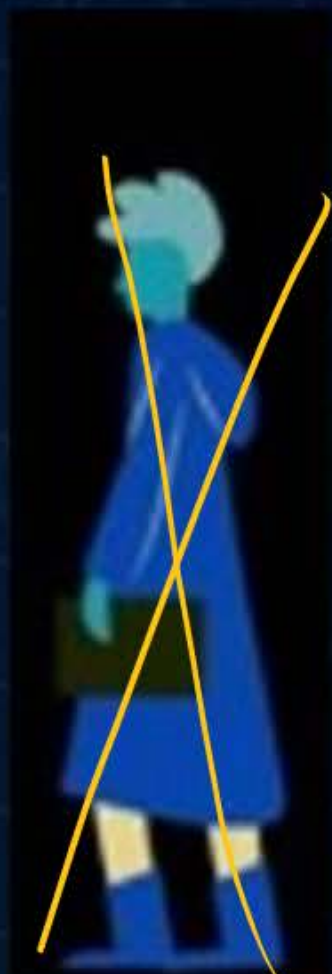
Queue



$f = 1$



Queue



$f=2$



$f=3$



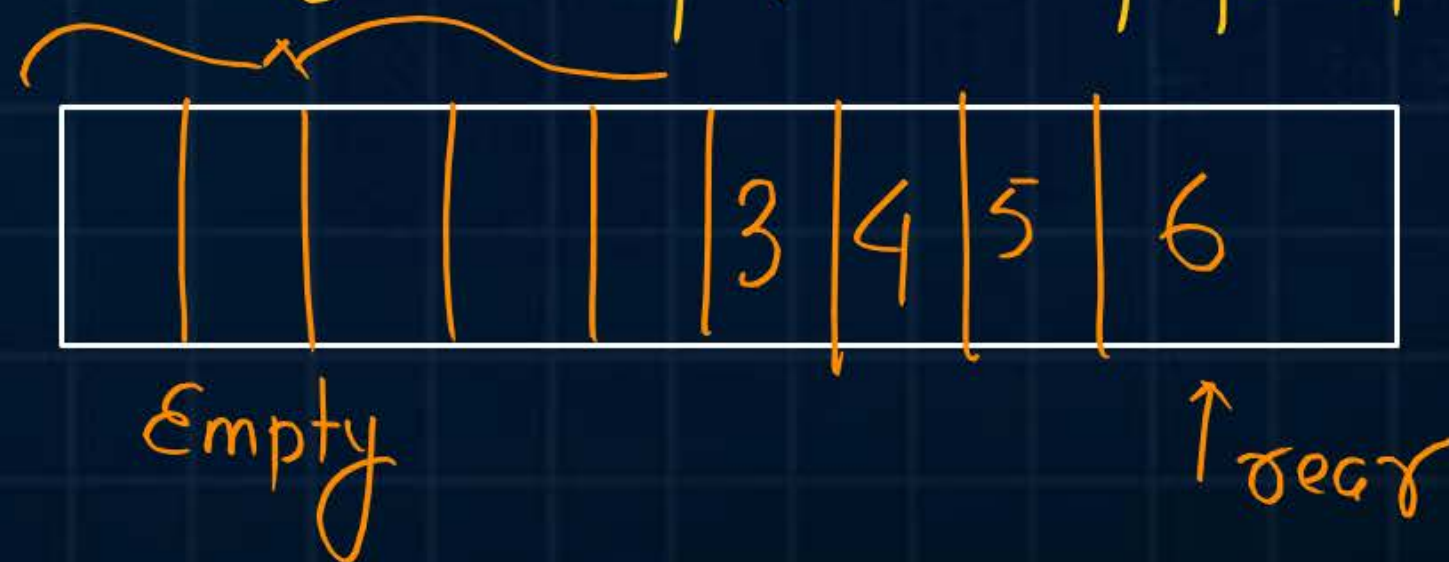


Queue

Linear queue : Empty queue front == -1

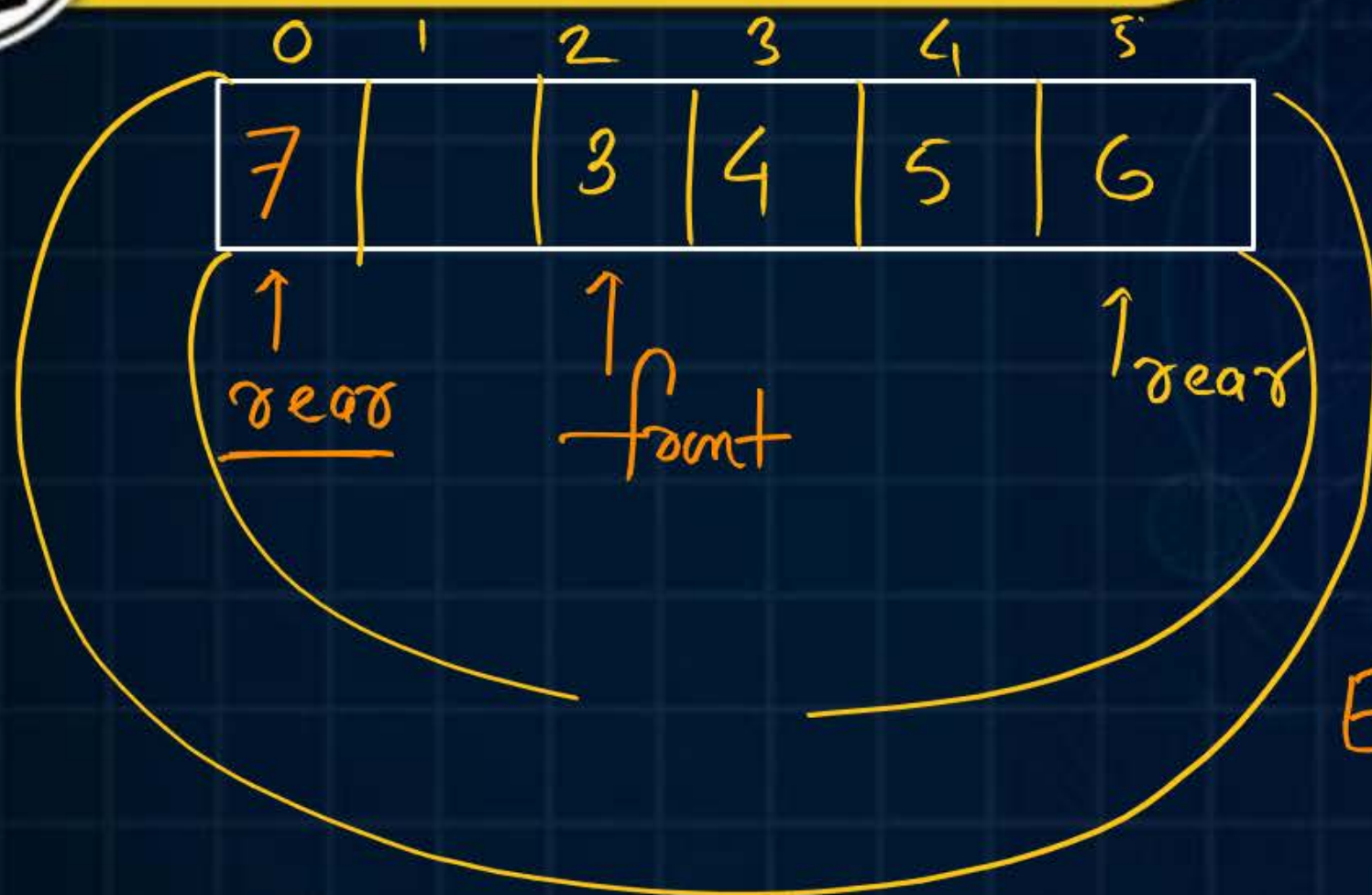
full : rear == Max - 1 ← (Index starts from

regardless of position of front





Circular Queue



$$\text{front} = (\text{front} + 1) \% \text{Max}$$

$$\text{rear} = (\text{rear} + 1) \% \text{Max}$$

$$\text{rear} = (5 + 1) \% 6 = \underline{0}$$

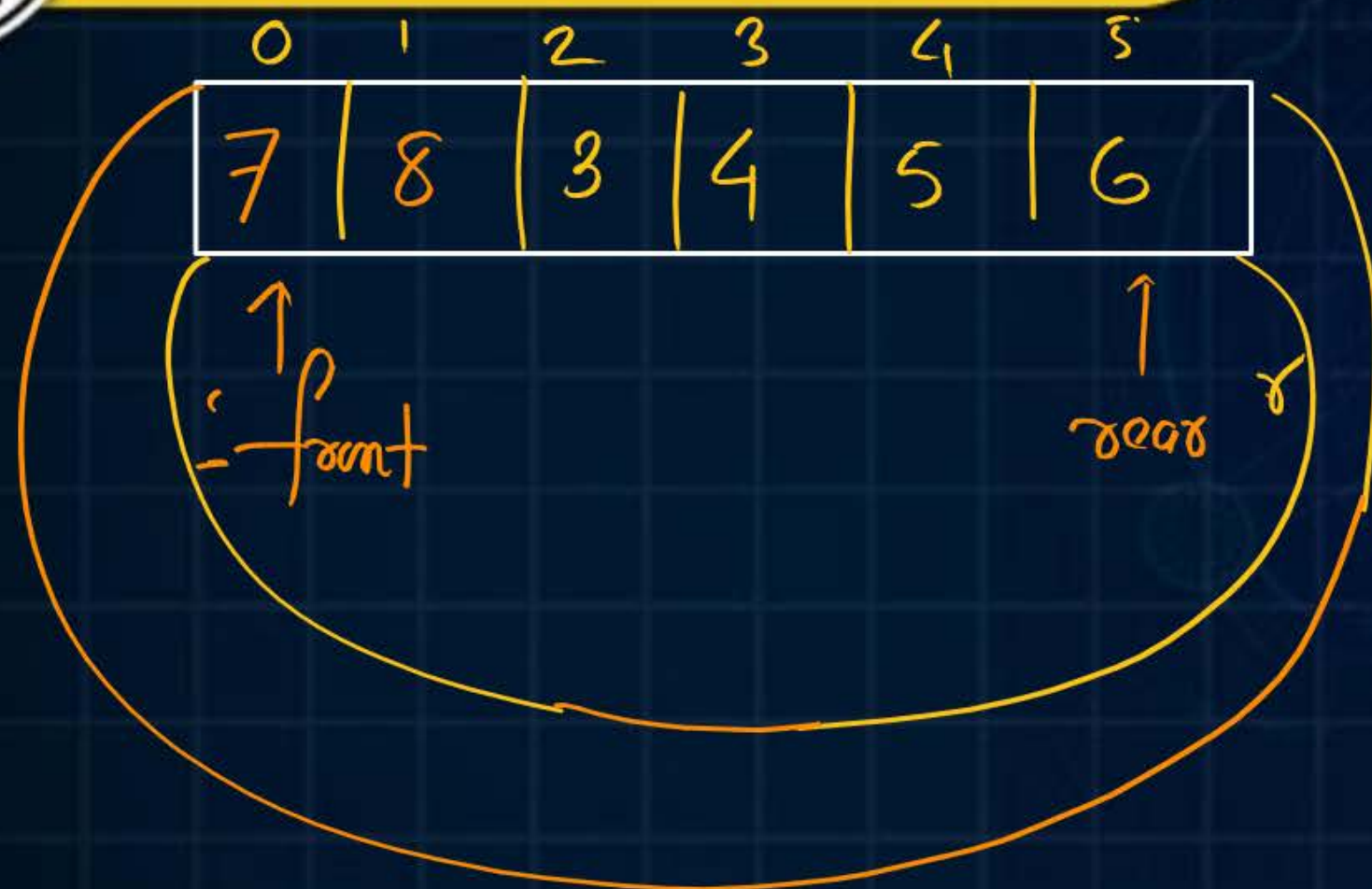
Insert 7

Circular Queue $\text{rear} < \text{front}$

Empty condition : $\text{front} == -1$



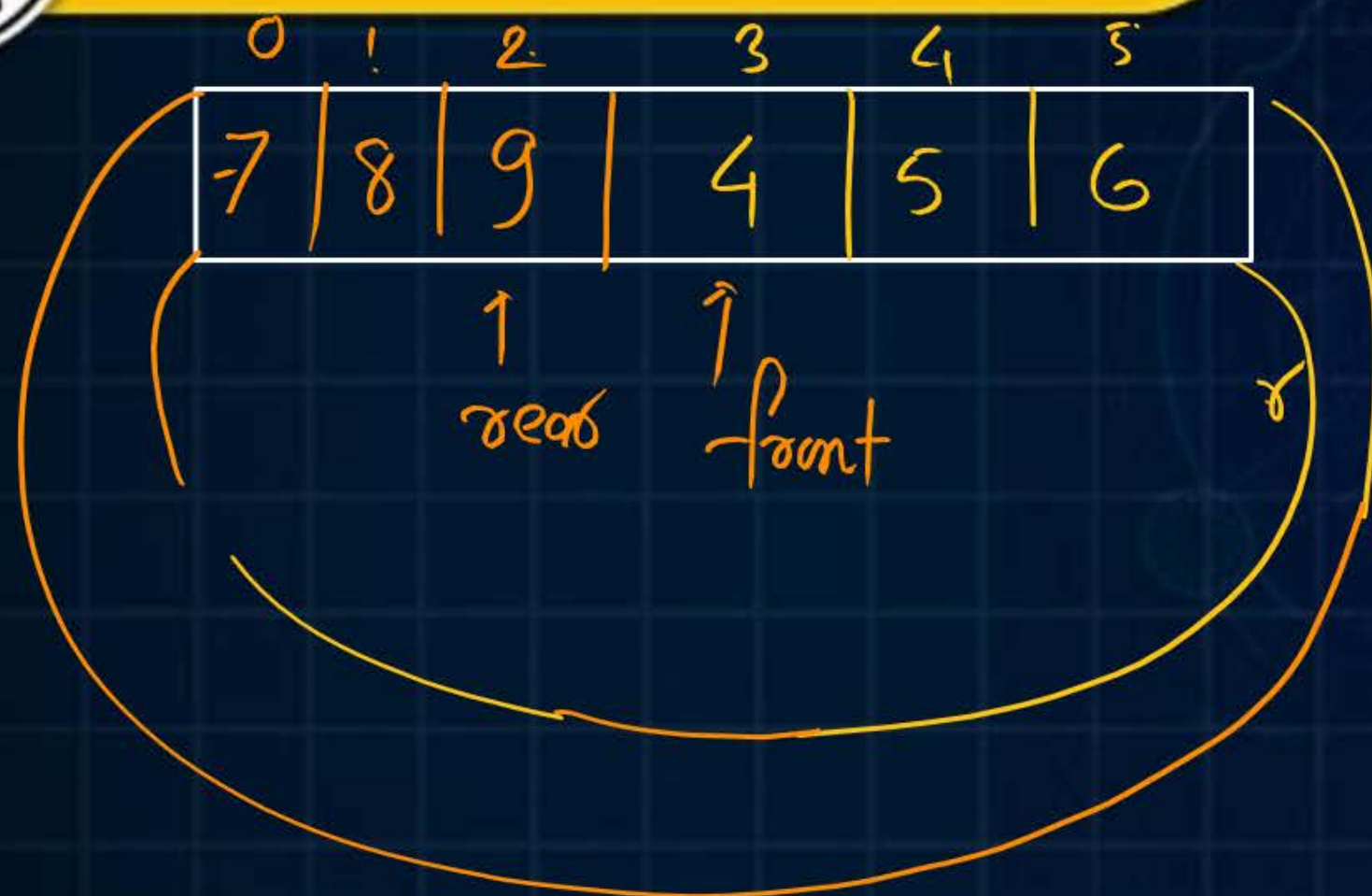
Circular Queue



$(front == 0 \ \&\& \ rear == Max - 1)$



Circular Queue



$$\text{front} == \text{rear} + 1$$

Queue is full

Enqueue(7)

Enqueue(8)

Enqueue(9)

$(front == 0 \text{ \&\& } rear == Max - 1) || (front == rear + 1)$ Max = 6

$f = 0, rear = 5$

$f = 3, r = 2$

$f = 4, r = 3$

$f = 2, r = 1$

$$front = (rear + 1) \% Max$$

Question

Consider the following sequence of operations on an empty stack: push(41); push(37); pop(); push(48); push(59); x = pop();

$$x = 59$$

$$59 + 26 = 85$$

59
37 48
41

Consider the following sequence of operations on an empty queue:

enqueue(18); enqueue(26); dequeue(); enqueue(35);

$$y = \text{dequeue} = 26$$

enqueue(44); y = dequeue(); The value of x + y is _____.

18	26	35	44				
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Topic: Question

Circular Queue

Q. If the Max is declared as 10 then what is the front and rear index after following operation



I: Enqueue(1), Enqueue(2), Enqueue(3) Enqueue(4) Enqueue(5) | Dequeue(),
Dequeue(), Dequeue() $f=3, r=4$ $f=0, r=4$

II: Enqueue(6), Enqueue(7), Enqueue(8) Enqueue(9) Dequeue(), Dequeue()
Enqueue(10), Enqueue(11), Enqueue(12)

if after Operation I front and rear value is w, x and After completion of II the front and rear value is y and z then value of $w+10x-2y+4z$ is front=5, rear=1

$$3+40-10+4 = \textcircled{37}$$

Question

For a given input string "XBBXDXXZX". The value return by the procedure

```
int counter(char *S){  
    int counter = 0;  
    char c;  
    while(*s) {
```

```
        c = *s;  
        s = s+1;  
        Enqueue(Q,c);
```

```
    }  
    if (! EmptyQueue(Q)) {  
        c = dequeue(q);  
        counter = 1;  
        While(!EmptyQueue(Q)){  
            if(c == dequeue(q))  
                counter++;  
        }  
    }
```

```
    }  
    return counter;  
}
```

final value is (5)



c = dequeue
c = x

frequency of
first character

Insert
char in queue

end of string

c = x

Empty queue

[NAT]

#Q. Consider the following circular Queue of size 10, implemented in C language having base address 1000 and size is 4 Bytes

	^{front}	²	³	⁴	⁵		⁷		
11	23	34	34	36	43	<u>16</u>	87 ₂₀	49 ₂₉	54 ₇₀ ← 9
	↑	↑ front							

Index start from 0, front index is at 7 and rear Index is at 2.

The following operation is implemented

Enqueue (34) Enqueue (36), Enqueue (43), dequeue(), dequeue(), dequeue(), dequeue() Enqueue (16), enqueue(20), Enqueue (29), Enqueue (70), x = Dequeue().

If after completion of the above operation y is the index of front and z is the index of rear then $x + y - z$ is 16.

$$x = 23 \quad : \quad \text{front} = 2 \quad \text{rear} = 9$$

$$23 + 2 - 9 = \boxed{16}$$

Q. Suppose a stack implementation supports an instruction REVERSE, which reverses the order of elements on the stack, in addition to the PUSH and POP instructions. Which one of the following statements is TRUE with respect to this modified stack?

- ~~(A)~~ A queue cannot be implemented using this stack.
- (B) A queue can be implemented where ENQUEUE takes a single instruction and DEQUEUE takes a sequence of two instructions. ~~X~~
- (C) A queue can be implemented where ENQUEUE takes a sequence of three instructions and DEQUEUE takes a single instruction.
- ~~(D)~~ A queue can be implemented where both ENQUEUE and DEQUEUE take a single instruction

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Stack implementation of
queue

Stack: LIFO

Queue:- FIFO

Queue: Simulation
using stack

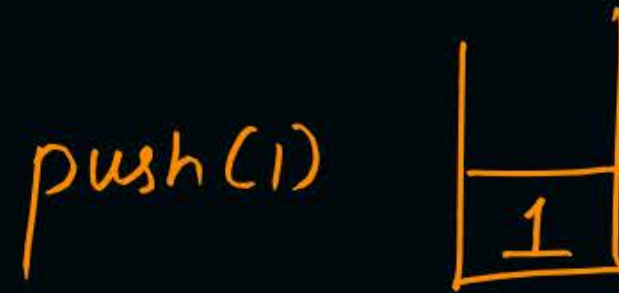
push
pop

Deque = pop

Enqueue : Reverse

push

Reverse



push(2)



Reverse



push



Reverse

push(1), push(2), push.

3
2
1✓

1
2
3

push = Enqueue
dequeue = Reverse
pop

Enqueue(4)
Enqueue(5)
dequeue

5
4
2
3

Reverse
pop

3
2
4
5

return 3

Question

Suppose the stack S support the operation of push pop and Reverse. Reverse operation reverse the entire content of stack. Similarly Queue support the operation of Enqueue , dequeue and reverse(reverse entire content of the queue. In addition to that it also support the operation of isEmpty() for stack and queue which returns true if stack or queue is empty.

Consider the following pseudo code

```
int dosomething () {
    int i,x,y;
    for (i = 1;i<=10;i++) {
        if ( i%2 == 0){
            reverse(s);
            push(s,i);
            reverse(s);
        }
        else {
            reverse(q);
            enqueue(Q,i);
            reverse(q);
        }
    }
    x=pop(s);
    y=dequeue(Q);
    return x+y;
}
```

Question

#Q. What is the output of the following code segment? S is the stack and Q is the Queue. Stack support push and pop operation and queue support enqueue and dequeue operation.

```
int a[50];
int n = 12;
for (int i = 100; i >= 2; i -= 2)
    a[i/2-1] = i;
for (int i = 0; i < 50; i++){
    S.push( a[i]);
    Q.enqueue(a[i]);
}
for (int i = 0; i < 4; i++){
    n = n + s.pop() + s.dequeue();
}
for (int i = 0; i < 2; i++){
    n = n - s.pop() - s.dequeue();
}
printf ("%d", n );
```

which of the following is TRUE for the above program?

A

The final value of n is 216 ✓

B

The top element of the stack is 14

C

The head of the queue is 88

D

The remaining element of the stack is 40

Question

#Q. What is the output of the following code segment? S is the stack and Q is the Queue. Stack support push and pop operation and queue support enqueue and dequeue operation.

```
int a[50];
int n = 12;
for (int i = 100; i >= 2; i -= 2)
    a[i/2-1] = i;
for (int i = 0; i < 50; i++) {
    S.push(a[i]);
    Q.enqueue(a[i]);
}
for (int i = 0; i < 4; i++) {
    n = n + s.pop() + s.dequeue();
}
for (int i = 0; i < 2; i++) {
    n = n - s.pop() - s.dequeue();
}
printf("%d", n);
```

$$a[49] = 100$$

$$a[48] = 98$$

$$a[0] = 2$$

100
98
96
94
6
4
2

2	4	6	8	...	100
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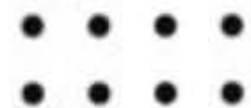
top = 88
head = 14

$$12 + 100 + 98 + 96 + 94 + 2 + 4 + 6 + 8 = 420$$

$$420 - 92 - 90 - 10 - 12 = 216$$

The word 'Thank' is written in a large, yellow, cursive script. A yellow arrow starts from the top of the 'T', extends horizontally to the right, and then curves downwards to point at the end of the word.

THANK



Keep Hustling!